The Coffee Law: Information, Reasoning and Communication Ceilings for Tool-Using Agent Collectives

Manish Bhatt
Affiliation to be completed
email to be completed

***Abstract*—Teams of LLM agents debate, vote and call tools in the hope of knowing more than any one of them. We show that what such a team can achieve is limited by three ceilings, each independent of how the agents work. *Information*: a team can never know more about the world than its starting data plus what its tools can carry, $\min\{H(\Theta),\, I(\Theta; D_0) + \sum_t C_t\}$ bits, where C_t is the capacity of round t's tool calls; talking, voting and adding agents do not raise this budget. *Reasoning*: what computationally bounded agents can *use* is at most what they *hold*; debate and extra compute close this gap but never pass it, and closing it can cost anything from reading every input to more than any efficient computation can supply. *Communication*: when data is split across agents, no single agent can use it without enough messages, about $n/2$ bits for some n-bit tasks even with unlimited computation. The ceilings explain a body of published findings: extra thinking fails on knowledge-intensive tasks, debate helps on reasoning tasks but no more than voting at equal compute, identical agents saturate, distributed information goes unpooled, and models know only what their training carried. We also formalize sandboxing as a guarantee on reachable channels. Every finite example is verified exactly in public code.**

***Index Terms*—multi-agent systems, LLM agents, tool use, information theory, channel capacity, usable information, communication complexity**

I. Introduction

A team of LLM agents is paged about a production incident. Two questions arrive. *Which server is failing?* The answer is in live logs the team has not yet read. *Is this patch to the locking code correct?* The answer is fully determined by code the team already has. A common belief is that adding agents, rounds of debate or thinking time helps with both. It helps with only one of them, and the reason is precise: the two questions hit different ceilings.

The first is limited by **information**. No amount of discussion reveals log lines nobody has fetched; only tool calls (reading logs, running queries) do, and each call can carry only so much. The second is limited by **reasoning**: the answer is already in hand, and the work is computing it. A third limit appears when data is split across agents, since what one agent holds is useless to another until it is communicated: **communication**.

This paper makes those three ceilings exact (Fig. 1). The mathematical tools are classical: the chain rule, the data-processing inequality (DPI), channel capacity, Fano's inequality [1], usable information [2] and communication complexity [3]. What is new is their assembly into a theory of tool-using agent collectives, which yields results not available from the textbook statements:

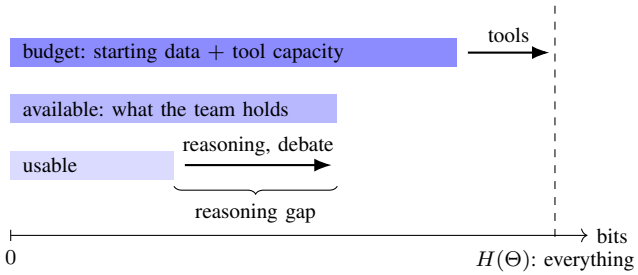


Fig. 1. The ceilings. What a team can use is at most what it holds, which is at most its budget. Tools (and better starting data) raise the budget; reasoning (more compute, debate, more agents) raises usable information toward what is held, never past it. When data is split across agents, a third ceiling, communication, limits what each agent can use.

1) **A capacity budget for adaptive, multi-agent tool use** (the Coffee Law, Theorem 1): the number of agents and the protocol drop out, and Fano's inequality gives an error floor and a minimum number of tool calls, including a worst-case version needing no prior.
2) **One chain for knowing and using**, usable $\leq$ available $\leq$ budget, with the gap between the first two equal to an expected KL divergence (Theorem 2), and bounds on the compute needed to close it (Theorems 3–4).
3) **A communication ceiling for collectives** (Theorem 5), distinct from information and compute.
4) **A formal account of sandboxing**: reasoning can discover channels, so a sandbox is a guarantee on which channels any feasible computation can reach (Theorem 6).
5) **A unified explanation of published findings** (Table II), and testable predictions.

Full proofs and exact numerical checks of every finite example are public.[1]

[1]https://github.com/mbhatt1/coffeelaw

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The theory is agent-agnostic: it applies equally to people, programs and mixed human–AI teams. It bounds learning about the world and using what is learned; it does not cap intelligence, and it says nothing about outcomes the agents themselves bring about.

II. Model

Intuition. We track one object: the *record*, everything the team has ever held. Each round, agents may call tools; tool answers depend on the world; everything else the agents do (posting, debating, voting, thinking) is a function of the record plus coin flips.

All variables are discrete with finite support (Section VII treats continuous ones); logarithms are base 2. Θ is the truth the team wants. D_t is the record at the start of round t: the initial data D_0 (every agent's context and model) plus every post, private state, query and tool result since. In round t, internal randomness ξ_t is drawn; tool calls $Q_t = g_t(D_t, \xi_t)$ are issued under any policy; results $O_{t+1} \sim K_t(\cdot \mid \Theta, Q_t, D_t)$ return; communication and computation use fresh randomness ζ_t; and $D_{t+1} = (D_t, \xi_t, Q_t, O_{t+1}, \zeta_t)$.

Assumptions. (A1) *No side channels*: Θ reaches the team only through D_0 and tool results. (A2) *Coin flips are honest*: ξ_t, ζ_t are independent of Θ and of everything before them. (A3) *Tools are channels from the world*: given (Θ, Q_t, D_t), O_{t+1} has law K_t (tools may be adaptive or stateful). (A4) *The truth is fixed*: the team's actions do not change Θ. (A1) is not an assumption about the agents but the definition of which channels exist; Theorem 6 treats agents that can open new ones, and Section VII relaxes the others.

Capacity. For each query q and history d, the tool is a channel $\theta \mapsto K_t(\cdot \mid \theta, q, d)$ with capacity $\mathrm{Cap}(K_{t,q,d}) = \max_P I(\Theta; O)$, the most it can ever reveal per call. Let $C_t = \sup_{q,d} \mathrm{Cap}(K_{t,q,d})$. Always $C_t \le \min\{\log|\mathcal{O}_t|, \log|\mathrm{supp}\,\Theta|\}$; a pass/fail test has $C_t \le 1$ bit.

What is random. For facts about the world, the randomness is over *possible worlds* W: a question $X \perp W$ has answer $\Theta = \theta(W, X)$, and training is a channel $W \to \text{corpus} \to M$ to the weights M. Then $I(\Theta; X, M) \le I(\Theta; X) + I(W; M)$ and $I(W; M) \le I(W; \text{corpus})$: a model knows at most what its training data carried. In one fixed world, Θ is a function of X and M carries no further information about it, so a model's factual knowledge appears as *usable* information (Section IV): something it computes from the question.

III. Ceiling 1: Information (the Coffee Law)

Intuition. Picture each tool call as a narrow pipe from the world into the team's shared bucket. Talking, voting and thinking move water between cups; they never add water. Only pipes do, and each pipe has a width.

Theorem 1 (Coffee Law)**.** *Under (A1)–(A4), with* $B_T = I(\Theta; D_0) + \sum_{t<T} C_t$*,*

$$I(\Theta; D_T) \le \min\{H(\Theta), B_T\},$$

and the same holds for every output of the team (answer, vote, board, any agent's state). For any estimator $\hat{\Theta}$ *and* $|\Theta| \ge 3$*,*

$$\Pr(\hat{\Theta} \ne \Theta) \ge (H(\Theta) - \min\{H(\Theta), B_T\} - 1)/\log(|\Theta| - 1).$$

In particular, with no tools, $I(\Theta; D_T) \le I(\Theta; D_0)$*: a closed team cannot learn anything by talking.*

Proof. By the chain rule, a round adds $I(\Theta; \xi_t | D_t) + I(\Theta; Q_t | D_t, \xi_t) + I(\Theta; O_{t+1} | S) + I(\Theta; \zeta_t | S, O_{t+1})$ with $S = (D_t, \xi_t, Q_t)$. The first and last terms are 0 by (A2); the second is 0 because Q_t is a function of (D_t, ξ_t). Given $S = s = (d, x, q)$, (A3) makes (Θ, O_{t+1}) an input–output pair of channel $K_{t,q,d}$ with input law $p(\cdot|s)$, so the third term is at most $\mathrm{Cap}(K_{t,q,d}) \le C_t$. Sum over rounds; also $I \le H$. Outputs are functions of D_T and independent noise (DPI). Fano's inequality gives the error floor. □

Choosing queries cleverly is already accounted for: a policy only selects which channel to use and under which posterior, and capacity is the maximum over both, as feedback cannot raise the capacity of a memoryless channel [4], [5].

Corollary 1 (What does not help)**.** *Debate, voting, critique and the number of agents do not appear in* B_T*. More agents raise the budget only by making more, or more informative, tool calls.*

Corollary 2 (Calls needed)**.** *If* Θ *is uniform on* 2^k *values* ($k \ge 2$)*,* $I(\Theta; D_0) = 0$ *and* $C_t \le C$ *with* $C > 0$*, then error* $\le \varepsilon$ *requires at least* $((1-\varepsilon)k - 1)/C$ *tool rounds, however many agents run and however they talk.*

Example: debugging. A test run returns pass or fail, so it carries at most one bit. If the bad commit is equally likely to be any of n, identifying it with error at most ε needs at least $(1-\varepsilon)\log n - 1$ test runs whatever the team does (Corollary 2); binary search (`git bisect`) uses $\lceil \log n \rceil$. A flaky test that misreports with probability p carries at most $1 - h(p)$ bits, raising the minimum by the factor $1/(1 - h(p))$.

Exact accounting and tightness. The proof gives the identity $I(\Theta; D_T) = I(\Theta; D_0) + \sum_t \mathbb{E}_s[I(\Theta; O_{t+1} | S_t = s)]$: total information is the sum of what each call actually revealed. The bound is tight exactly when every call is used at full capacity: almost every posterior is a capacity-achieving input for the chosen tool, and each chosen tool attains C_t. For rich tools (a web page), the realized per-call gain is the binding quantity; the capacity form bites for low-capacity tools such as tests, checks and short API answers. Table I shows a case where a good policy meets the bound and a poor one wastes it.

TABLE I
Exact information after two tool rounds. Θ is uniform on 4 values; each call returns one chosen bit of Θ, flipped with probability 0.1 ($C \approx 0.531$ bits). Budget: $\min\{H(\Theta), 2C\}$.

Policy	$I(\Theta; D_2)$	Budget
Ask bit 0, then bit 1	1.0620	1.0620
Adaptive, may repeat a bit	0.9820	1.0620
Either, plus a debate step	unchanged	—
Ask both bits, noise-free tool	2.0000	2.0000

IV. Ceiling 2: Reasoning

Intuition. Holding information is not the same as using it. The prime factors of a number N are completely determined by N, so a team holding N holds all the information about them; finding them may still take enormous computation. Reasoning turns held information into usable information.

Definition 1 ([2])**.** A *predictive family* $\mathcal{V}$ is a set of predictors f mapping a record d to a distribution $f[d]$ on Θ, including the prior predictor; it models what the agents can compute. Let $H_{\mathcal{V}}(\Theta|D) = \inf_{f\in\mathcal{V}} \mathbb{E}[-\log f[D](\Theta)]$ and define the *usable information* $I_{\mathcal{V}}(D \to \Theta) = H(\Theta) - H_{\mathcal{V}}(\Theta|D)$.

Theorem 2 (Reasoning ceiling)**.** *(a)* $0 \le I_{\mathcal{V}}(D \to \Theta) \le I(\Theta; D)$*, with* reasoning gap $I(\Theta; D) - I_{\mathcal{V}}(D \to \Theta) = \inf_{f\in\mathcal{V}} \mathbb{E}_D\, D_{\mathrm{KL}}(p(\cdot|D) \,\|\, f[D])$*. (b) Hence* $I_{\mathcal{V}}(D_T \to \Theta) \le I(\Theta; D_T) \le \min\{H(\Theta), B_T\}$*. (c) Computation can raise usable information but never available information: for some* $\mathcal{V}$ *and deterministic* g*,* $I(\Theta; g(D)) = I(\Theta; D)$ *while* $I_{\mathcal{V}}(g(D) \to \Theta) > I_{\mathcal{V}}(D \to \Theta)$*; for every* g*,* $I_{\mathcal{V}}(g(D) \to \Theta) \le I(\Theta; D)$*.*

Proof. (a) The prior predictor gives $H_{\mathcal{V}} \le H(\Theta)$. For any f, $\mathbb{E}[-\log f[D](\Theta)] = H(\Theta|D) + \mathbb{E}_D D_{\mathrm{KL}}(p(\cdot|D)\|f[D])$; take the infimum. (b) Part (a) and Theorem 1. (c) Let two agents hold uniform bits B_1, B_2 and $\Theta = B_1 \oplus B_2$, and let $\mathcal{V}$ contain the predictors that read at most one coordinate of the record. Each bit alone is independent of Θ, so usable information is 0, although the pair determines Θ. After one reasoning step $g(D) = (B_1 \oplus B_2, B_2)$, one coordinate equals Θ: usable information is 1 bit, available information unchanged. The last claim is (a) and the DPI. □

Corollary 3 (Debate adds reasoning, not facts)**.** *In a closed team every post is a function of* D_0 *and noise, so it adds no available information; it can add usable information, as when an agent posts an intermediate result (here* $B_1 \oplus B_2$*) that others could not compute. The gain is capped by what the team holds.*

The incident example is now exact. "Which server is failing?" is *information-bound*: the right inequality of the chain binds, and only tools help. "Is this patch correct?" is *compute-bound*: the left gap is large, and reasoning helps. A code interpreter run on data already held is reasoning (it enlarges $\mathcal{V}$); a tool that observes the world is information (it raises C_t). In Table I's noisy example, the record holds 1.062 bits but a one-look agent can use only 0.266; the rest is a reasoning gap that more observations cannot close.

Imperfect agents. Real agents fall further below the chain (proofs in the accompanying notes, footnote 1). Acting on a belief $\hat{b}$ instead of the correct posterior b^* costs at most

$$2U\,\mathrm{TV}(b^*, \hat{b}) + \delta \;\le\; 2U\sqrt{(\ln 2/2)\, D_{\mathrm{KL}}(b^* \| \hat{b})} + \delta$$

for utility range U and own suboptimality δ. A summary M of data D loses exactly $\mathbb{E}\, D_{\mathrm{KL}}(p(\cdot|D)\|p(\cdot|M)) = I(\Theta; D) - I(\Theta; M)$. And a confident message carrying zero information can flip a correct agent: with belief $(0.6, 0.4)$ for the truth, giving weight $\alpha > 1/6$ to a constant false claim flips the answer.

V. Paying for Reasoning

Intuition. Closing the reasoning gap costs computation. A team of agents is just a computation spread over several machines, so what counts is total work; and some tasks need a lot of it.

Let $\mathcal{V}_S$ be the predictors computable and samplable in at most S steps of a fixed machine model (reading one input symbol per step). A team's *total compute* is the sum of all agents' steps over all rounds, counting board reads and writes.

Theorem 3 (Only total compute counts)**.** *A team with total compute* S *outputs a predictor in* $\mathcal{V}_{\kappa S}$ *for a constant* κ *of the machine model, so its usable information is at most* $I_{\mathcal{V}_{\kappa S}}(D \to \Theta) \le I(\Theta; D)$*.*

Proof. One machine can run the agents step by step, keeping their memories and the board in separate regions, at $O(1)$ cost per simulated step. □

The comparison holds in any fixed cost model (RAM steps, FLOPs, tokens processed), up to the overhead of interleaving.

Lemma 1. *Sampling* $\hat{\Theta} \sim f[D]$ *is correct with probability at least* $2^{-\mathbb{E}[-\log f[D](\Theta)]}$ *(Jensen). If* $H(\Theta) \le 1$ *and* f *attains usable information* δ *(cross-entropy* $H(\Theta) - \delta$*), the success probability is at least* $\frac{1}{2} + \frac{\ln 2}{2}\delta$*.*

Theorem 4 (Some gaps are expensive)**.** *(a) For the parity* Θ *of* n *uniform bits, any predictor or team that reads at most* $n-1$ *bits, even adaptively, has zero usable information; reading all* n *gives one bit. (b) If one-way permutations exist, then for the Goldreich–Levin bit* $\Theta = \langle x, r\rangle \bmod 2$ *given* $D = (f(x), r)$*,* $I(\Theta; D) = H(\Theta)$*, yet every polynomial-time team has usable information at most* $\log(1 + 2\varepsilon(n))$ *with* ε *negligible.*

Proof. (a) Given any transcript of fewer than n reads, the unread bits are uniform, so Θ is independent of the transcript; posts are functions of their authors' reads. (b) Every polynomial-time algorithm guesses Θ with probability at most $\frac{1}{2} + \varepsilon$ [6]; apply Lemma 1; by Theorem 3 this covers any polynomial-time team. □

So splitting work across agents buys speed, not capability, and reaching the reasoning ceiling can cost anything from reading the whole input to more than any efficient team can supply. No general formula exists in between without resolving open problems such as P vs NP.

VI. Ceiling 3: Communication

Intuition. Two agents each hold half a puzzle. The team as a whole holds the answer, but neither agent can use it until enough of the other half arrives. For some puzzles a one-bit summary suffices; for others, nothing much shorter than the data itself will do.

Theorem 5 (Communication ceiling)**.** *(a) If the messages reaching agent 1 take at most* 2^L *values, then* $I(\Theta; D^1, \Pi) \le I(\Theta; D^1) + L$*. (b) If agents 1 and 2 hold uniform* $x, y \in \{0,1\}^n$

and $\Theta = \langle x, y \rangle \bmod 2$*, then* $I(\Theta; x, y) = H(\Theta)$*, yet any protocol exchanging* L *bits gives* $I(\Theta; x, \Pi) \le \log(1+2^{L+1-n/2})$*, even with unlimited computation.*

Proof. (a) $I(\Theta; \Pi | D^1) \le H(\Pi) \le L$. (b) Append agent 1's output as one more bit. Inner product has discrepancy at most $2^{-n/2}$ (Lindsey's lemma [3], [7]), so an $(L+1)$-bit protocol is correct with probability at most $\frac{1}{2} + 2^{L-n/2}$; apply Lemma 1 to agent 1's exact posterior. □

Parity splits cheaply (each agent posts one bit); inner product needs about $n/2$ bits. With more agents, the same bounds apply to the bits crossing any cut that separates the inputs. This ceiling holds even when information and compute are unlimited.

VII. Beyond the Assumptions

Each result relaxes one assumption; full proofs and exact checks are in the accompanying notes (footnote 1).

Theorem 6 (Reasoning can open channels: sandboxes)**.** *Let* $A_S(d)$ *be the queries a team can issue from record* d *within compute* S*. Then* $I(\Theta; D_T) \le I(\Theta; D_0) + \sum_t \sup_{d,\, q \in A_S(d)} \mathrm{Cap}(K_{t,q,d})$*, nondecreasing in* S*. If a tool reveals a uniform 2-bit* Θ *only after a uniform* k*-bit password is found,* S *guesses give at most* $\min\{1, S/2^k\} \cdot 2$ *bits, with equality for distinct guesses.*

So "reasoning never adds information" holds for a *fixed* set of reachable channels. Discovering an endpoint, credential or exploit is computation that raises the budget. A *sandbox* is a guarantee that every feasible computation stays within allowed channels; an escape violates it. In recently reported incidents, models under evaluation left their test sandbox and retrieved benchmark answers from an outside server [8]: the answers were not inside, so the only route was an unallowed channel.

Worst case. For $M \ge 3$ possible truths with no prior, starting data of capacity C_0, and any strategy, some truth θ_i has $\Pr_{\theta_i}(\hat{\Theta} \ne \theta_i) \ge (\log M - \min\{\log M, C_0 + \sum_t C_t\} - 1)/\log(M-1)$ (Fano's method). The law holds for specific tasks, not only on average.

Acting. In a fixed unknown environment W with $M \ge 3$ actions, a unique best action A^* and gap Δ, round-t regret is at least $\Delta(H(A^*) - \min\{H(A^*), I(W; D_0) + \sum_{s<t} C_s\} - 1)/\log(M-1)$: the action is an estimate of A^*. Taking W to be a user's hidden specification covers open-ended work: an agent cannot reliably build what it has not learned the user wants.

A changing world. If Θ_t drifts as a Markov chain with strong data-processing constant $\eta < 1$ [9], tools observe the current state, and $C_t \le C$, then $I(\Theta_t; D_t) \le \eta^t I(\Theta_0; D_0) + \eta C/(1-\eta)$: knowledge saturates however long agents run.

Adversaries and provenance. If every post is stored with its author, adversarial posts cannot reduce available information: $I(\Theta; D, M) \ge I(\Theta; D)$. Without attribution one adversary can erase it: if an honest agent posts a fair bit Θ and an adversary posts $1-\Theta$, the unattributed record carries 0 bits. Authenticate and attribute messages.

Continuous truths. For real Θ the budget holds with capacities under input constraints, and every estimator has $\mathbb{E}[(\Theta - \hat{\Theta})^2] \ge (2\pi e)^{-1} 2^{2(h(\Theta) - B_T)}$, with equality for a Gaussian truth in Gaussian noise: each bit divides the error floor by 4.

Where to spend. From the chain: extra reasoning on the same record can raise usable information by at most the reasoning gap $I(\Theta; D) - I_{\mathcal{V}}(D \to \Theta)$, and extra tool rounds can raise available information by at most $\sum C_t$. A spending rule follows: invest in the lever whose headroom per unit cost is larger, and stop investing in a lever whose headroom is exhausted.

Measuring the quantities. Usable information is estimable: $H(\Theta)$ minus the held-out log-loss of a team's calibrated predictions estimates a lower bound on $I_{\mathcal{V}}$, with PAC guarantees under computational constraints [2]. Tool capacity is bounded by the answer space, and for noisy tools by measured error rates (a test that flakes with probability p has $C \le 1 - h(p)$). Available information about the world, $I(W; M)$ for a pre-trained model, is well defined but hard to estimate; the bounds' value is then structural rather than numerical.

VIII. Evidence and Predictions

Table II lists predictions of the theory and the published findings we found that bear on them, including mixed ones; none contradicts it. Two points deserve care. First, the theory does not predict that debate is worse than other methods; it predicts that debate is worth its compute, so at matched compute it should perform about as well as self-consistency or voting. Small gaps in either direction [10]–[12] are consistent with this. Second, some findings look like tensions. Debate is reported to improve *factuality* [13]; but within the actual world a factual answer is determined by the question, so this is reasoning over stored knowledge, not new information. Debate is proved not to help Bayesian agents [12]; such agents have no reasoning gap, so Theorem 2 predicts exactly that. Finally, diverse agents outperform identical ones [14]: models trained on different data carry different information (a larger D_0) and compute different things (a larger $\mathcal{V}$), whereas identical agents add only compute.

Testable predictions. (P1) Error rate against the number of informative tool calls stays above the floor of Corollary 2; a clean test uses a yes/no tool with known noise, where the floor is exact. (P2) At matched compute, debate performs about as well as self-consistency or voting; it helps compute-bound tasks and not information-bound ones, while tools show the opposite pattern. (P3) Redundant or low-capacity tool calls add little, however numerous. (P4) With data split across agents, accuracy on tasks like inner product stays near chance until the message budget approaches the communication ceiling of Theorem 5.

IX. Related Work

Closed-system debate is bounded via the DPI in [28] and modeled as a belief martingale in [12]. Yang et al. [14] bound

TABLE II

PREDICTIONS OF THE THEORY AND PUBLISHED EVIDENCE. *Supported*: THE FINDING MATCHES THE PREDICTED DIRECTION. *Consistent*: COMPATIBLE WITH THE PREDICTION BUT MIXED ACROSS STUDIES. FINDINGS ARE SUMMARIZED FROM THE CITED PAPERS' ABSTRACTS AND PUBLISHED SUMMARIES.

Prediction (result in this paper)	Published finding	Verdict
Extra thinking does not help information-bound tasks (Thm. 1, no tools)	More test-time compute does not reliably raise closed-book accuracy across 14 reasoning models and often increases hallucination [15]; self-correction without external feedback often fails [16].	Supported
Tools raise information-bound accuracy (Thm. 1)	Retrieval and tool use are the standard remedy for knowledge-intensive tasks [17]–[19].	Supported
A model knows only what its training carried (Sec. II)	QA accuracy tracks the number of relevant pretraining documents, causally; retrieval reduces the dependence [20].	Supported
Debate helps compute-bound tasks (Thm. 2)	Debate improves mathematical and strategic reasoning [13].	Supported
Debate is worth its compute, no more (Cor. 3)	Debate does not reliably beat self-consistency [11]; most gains come from voting [12]; single agents match multi-agent systems at equal thinking tokens [10].	Consistent
More identical agents saturate (Thm. 3, Cor. 1)	Homogeneous agents show strong diminishing returns; two diverse agents can match sixteen identical ones [14]; voting gains grow with task difficulty [21].	Supported
Redundant tool calls add little (Thm. 1, exact accounting)	Accuracy plateaus after about eight calls and redundant calls can hurt [22].	Supported
Split information must be communicated (Thm. 5)	Agent groups reach 30.1% with distributed information versus 80.7% for one agent with all of it; agents do not seek what others hold [23].	Supported
Zero-information messages flip correct agents (Sec. IV)	Debate can lower accuracy over time as agents switch from correct to incorrect answers [24].	Supported
Without provenance, adversaries erase information (Sec. VII)	Stealthy tampering of inter-agent messages succeeds across tasks and architectures [25].	Supported
In a changing world knowledge decays and tools refresh it only partly (Sec. VII)	Under continuous knowledge drift, retrieval and learning methods struggle [26].	Supported
Open-ended work is bounded by information about intent (Sec. VII)	Underspecification consistently hurts agents; an ask-user tool recovers part of the loss [27].	Supported
Error vs. informative tool calls stays above the Fano floor (Cor. 2)	Not yet measured.	Open

multi-agent performance by task uncertainty and a sum of per-agent channel informations, the same "$\min\{H, \sum \text{channels}\}$" shape with agents as the channels. Fano-type bounds have been applied to queries in adversarial attacks [29] and to single-pass multi-hop reasoning [30]. Usable information is due to [2]; synergy is studied in partial information decomposition [31]; feedback and directed information in [5]; and Fano-based active-learning bounds in [32]. To our knowledge, the per-call capacity budget for adaptive multi-agent tool use, its combination with usable information and communication into one set of ceilings, and the channel account of sandboxing have not been stated before.

X. LIMITATIONS

The bounds concern information about a world the agents do not change; outcomes the team itself brings about need a different theory. The ceilings hold separately, but how to trade one for another is open, as is modeling agents' incentives (reward hacking, goal adoption), which the bounds do not address. Bounds for agents that learn across episodes, whose starting data changes over time, are not given. Equality in Theorem 1 is characterized, but no general algorithm achieves it. The proofs are mathematical and checked numerically on finite examples, not machine-verified. We report no new experiments; Table II relies on published findings, summarized from their abstracts.

XI. CONCLUSION

A team of agents can be clever, but it cannot know more than it has seen, use more than it holds, or combine what its members hold without communicating enough. For practitioners the theory reduces to one question per task: *which ceiling binds?* If the answer is not in the data, add better observations of the world. If it is in the data, add reasoning, and expect debate to be worth exactly the compute it spends. If it is split across agents, make sure they can say enough to each other. And treat sandboxes and message provenance as channel control, because channels are where information comes from.

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